EP-UNet++: Advancing Landslide Monitoring Through Enhanced Remote Sensing Semantic Segmentation

HaiYang Li, Jing Wang, Shuguang Wu, Yawei Wang, Yehemin Gao, and Guigen Nie

***Abstract*—Landslide deformation monitoring is crucial for mitigating geological hazards and ensuring public safety in disaster-prone regions. Remote sensing imagery offers unparalleled advantages in capturing large-scale, high-resolution terrain data for landslide monitoring, but its effectiveness is hindered by challenges in accurately segmenting complex terrain features and delineating boundaries. This study proposes EP-UNet++, a hybrid architecture that integrates UNet++, pyramid vision transformer (PVT), and EfficientNet through multilevel cross-model fusion, cross-resolution alignment, and 11-channel input adaptation. The framework effectively combines local and global features, substantially enhancing the accuracy and generalization of landslide remote sensing image segmentation. In addition, local interpretable model-agnostic explanations (LIMEs) analysis is employed to interpret the contribution of each input factor to the model's predictions. Experimental results show that the EP-UNet++ achieves a mean intersection over union (mIoU) of 74.15, a mean dice coefficient (mDice) of 85.16, and a specificity of 0.979. It achieves an improvement of roughly 4% in mIoU and 2% in mDice over the second-best model, Swin-UNet. The model also exhibits faster convergence during training and robust generalization for landslide pixel detection. These results highlight the EP-UNet++ as a highly effective approach for precise and reliable landslide remote sensing image segmentation, with potential benefits for cost-effective environmental monitoring and disaster mitigation.**

***Index Terms*—Deep learning, disaster prevention, EP-UNet++, landslide monitoring, remote sensing, semantic segmentation.**

I. Introduction

LANDSLIDE monitoring techniques can be broadly categorized into in situ monitoring [3], [4], [5], geophysical methods [6], [7], [8], and remote sensing [9], [10], [11]. In situ monitoring, including GNSS [12] and extensometers [13], provides high-precision data but is often limited by spatial coverage and high installation costs. Geophysical methods, such as ground-penetrating radar [14] and seismic monitoring [15], are effective in identifying subsurface instabilities but require specialized equipment and are constrained to localized areas. Remote sensing has emerged as a cost-effective and scalable solution for landslide monitoring. Techniques such as synthetic aperture radar (SAR) [16], [17], optical imagery [18], [19], and light detection and ranging (LiDAR) [20], [21] enable large-scale observations, capturing both pre-event and postevent landslide characteristics. Among these, remote sensing image analysis stands out for its ability to detect landslide-prone areas based on terrain features, vegetation changes, and soil displacement patterns. Its capability to integrate multitemporal and multispectral data further enhances its application in landslide detection and risk assessment [22]. Compared to traditional methods, remote sensing-based approaches provide a more comprehensive understanding of landslide dynamics, making them indispensable for proactive disaster management and decision-making.

The development of remote sensing imagery processing methods for landslide monitoring has evolved significantly over the years. Early approaches relied on manual interpretation of aerial and satellite imagery, which, while effective for small-scale studies, are time-consuming and subjective. With advancements in technology, automated techniques such as spectral thresholding, change detection, and terrain analysis became widely adopted, enabling broader spatial coverage and higher efficiency. The introduction of SAR and optical imagery has further enhanced monitoring capabilities, providing critical insights into surface deformation and land cover changes [23], [24], [25]. Recently, machine learning and deep learning methods, such as random forest [26], convolutional neural networks (CNNs) [27], and Transformer models [28], [29], have demonstrated remarkable accuracy in extracting complex spatial patterns and automating landslide detection. However, several challenges remain. Occlusions caused by dense vegetation, variability in landslide features across different terrains, and the computational complexity of processing high-resolution multitemporal datasets pose significant bottlenecks. Moreover, existing methods often struggle with balancing detection accuracy and scalability in real-time applications. Addressing these limitations requires further integration of

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advanced algorithms with multisource remote sensing data to develop robust, efficient, and adaptable solutions for landslide monitoring.

In the field of landslide remote sensing image recognition, EfficientNet, pyramid vision transformer (PVT), and UNet++ have emerged as influential models, each with distinct advantages and limitations. EfficientNet employs a compound scaling method to balance network depth, width, and resolution, achieving high accuracy with lower computational costs [30], [31]. However, its reliance on fixed grid convolution limits its ability to capture global contextual information, which is crucial in complex remote sensing tasks. PVT utilizes self-attention mechanisms to effectively capture global and hierarchical features, making it highly suitable for analyzing large-scale remote sensing images [32], [33]. Yet its high computational overhead and complexity in handling fine-grained details can hinder practical deployment. Originally developed for biomedical image segmentation, UNet++ features nested skip connections and a flexible architecture, enhancing its ability to achieve precise boundary delineation in remote sensing imagery [34], [35]. However, its performance can decline with large-scale or complex datasets due to its convolution-centric design. EfficientNet provides an efficient feature extraction mechanism suitable for high-resolution data processing, PVT excels in capturing global and hierarchical spatial features essential for large-scale analysis, and UNet++ ensures precise boundary refinement critical for detailed segmentation tasks. Conventional segmentation methods, though effective in general vision tasks, are often inadequate for landslide detection because they predominantly rely on single-source imagery, such as red–green–blue (RGB) bands [36], [37]. This reliance on RGB data alone limits the models' ability to distinguish landslides from spectrally similar surfaces, including bare soil, agricultural fields, or built-up areas, and reduces adaptability across diverse terrains. In addition, RGB-based approaches fail to fully capture critical environmental and topographic factors, such as slope, soil moisture, and vegetation cover, which are key drivers of slope instability [38]. Geospatial fusion overcomes these limitations by integrating multisource features such as spectral indices, topography, and environmental variables, enhancing feature discrimination and improving the robustness of landslide delineation in complex conditions.

This study makes several key contributions to advancing landslide monitoring based on remote sensing imagery.

1) It proposes an EP-UNet++ model that integrates EfficientNet, PVT, and UNet++ to achieve effective fusion of global and local features.
2) The model enhances adaptability to complex and large-scale remote sensing datasets, improving both segmentation accuracy and computational efficiency.
3) Extensive experiments demonstrate the stability, scalability, and robustness of EP-UNet++, providing a reliable framework for landslide detection in remote sensing imagery.

Although the study is constrained by the availability of specific datasets, the proposed EP-UNet++ provides a promising and generalizable framework for accurate and efficient landslide detection in remote sensing applications.

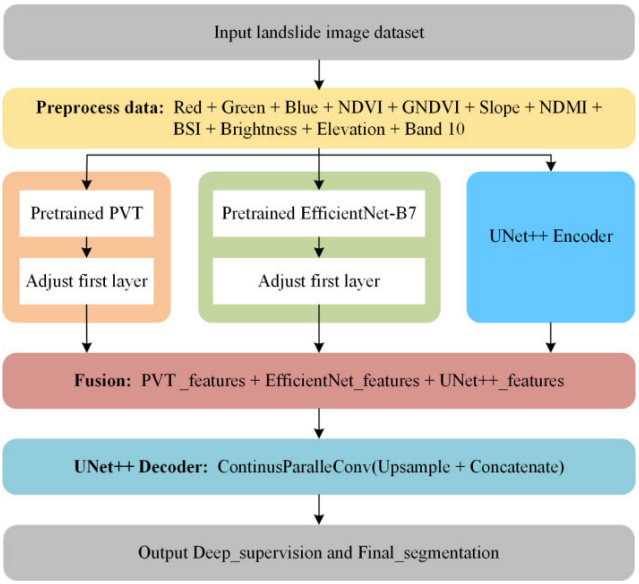


Fig. 1. Flowchart of the EP-UNet++ for landslide monitoring using multimodal remote sensing imagery.

The structure of this article is organized as follows. Section II presents the methodology, detailing the proposed EP-UNet++, including its architectural design and implementation for landslide monitoring. Section III describes the experiments and results, covering the dataset, evaluation metrics, and comparative analysis to validate the effectiveness of the proposed method. Section IV provides a discussion of the findings, emphasizing the strengths, limitations, and potential applications of the model in real-world scenarios. Finally, Section V concludes the article with a summary of key contributions and suggestions for future research directions.

II. Methodology

In this study, a hybrid strategy is adopted to develop a robust and efficient framework for landslide monitoring using remote sensing imagery. As shown in Fig. 1, the study starts with multimodal remote sensing imagery, followed by preprocessing to extract key features, including RGB bands, normalized difference vegetation index (NDVI), green NDVI (GNDVI), slope, normalized difference moisture index (NDMI), bare soil index (BSI), brightness, elevation, and thermal infrared Band 10. These features are first normalized using an adaptive strategy based on the maximum value of each feature, and are then equally fed into three pretrained models: PVT, EfficientNet-B7, and UNet++ encoder for feature extraction.

Since the default architectures of PVT and EfficientNet-B7 are designed for three-channel inputs, their input layers are adjusted to accommodate the 11-channel data. The features extracted by these models are subsequently fused to leverage their complementary strengths. The fused features are passed through the UNet++ decoder module for upsampling and refinement. The methodology ensures multiscale feature representation and precise segmentation, producing supervisory outputs for each layer and the final landslide monitoring results. The selected features comprehensively characterize landslide-prone environments. RGB bands provide spectral

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information for surface characterization, while NDVI and GNDVI indicate vegetation cover and stability. NDMI reflects soil moisture, and BSI highlights exposed soil surfaces linked to instability. Brightness supports surface type discrimination, elevation, and slope quantify terrain conditions, and thermal infrared Band 10 captures land surface temperature influencing moisture and weathering processes. The research process begins with data preprocessing to ensure high-quality inputs for model training. The proposed EP-UNet++ integrates EfficientNet for feature extraction, PVT for global context modeling, and UNet++ for precise segmentation. The proposed model is trained and validated on benchmark datasets, with performance evaluated against state-of-the-art methods. By leveraging the strengths of advanced deep learning architectures, this approach combines efficiency, global feature representation, and fine-grained segmentation to address the challenges in landslide detection effectively.

A. Feature Extraction

Effective feature extraction is fundamental for achieving high performance in segmentation tasks, particularly when dealing with multimodal remote sensing images. This study employs a combination of pretrained EfficientNet-B7, pretrained PVT, and U-Net++ encoders to extract diverse and complementary features. Each model contributes unique strengths: EfficientNet-B7 provides computational efficiency and multiscale feature representation, PVT captures long-range dependencies and contextual information, and U-Net++ improves feature granularity with its densely connected encoder pathways, facilitating more effective hierarchical feature extraction. Together, these models form a robust feature extraction framework, ensuring that both spatial and semantic information are preserved for downstream decoding.

1) Pretrained EfficientNet-B7 (Optimizing Scalable and Precise Local Feature Extraction): EfficientNet-B7 is a deep learning architecture designed to achieve a balance between computational efficiency and model accuracy. Developed as part of the EfficientNet family, it introduces a compound scaling strategy that simultaneously adjusts the depth, width, and resolution of the network in a principled manner [48]. This approach ensures that the model maximizes its representational capacity without unnecessary computational overhead. Built upon mobile inverted bottleneck convolution (MBConv) blocks and enhanced with squeeze-and-excitation (SE) modules, EfficientNet-B7 is capable of extracting rich feature representations while maintaining scalability. Its ability to achieve state-of-the-art performance with reduced parameters and floating-point operations per second makes it particularly suitable for tasks requiring efficient and accurate feature extraction.

Fig. 2 illustrates the workflow of EfficientNet-B7, highlighting its key architectural components and the compound scaling strategy. The model begins with an input layer, which undergoes a series of MBConv blocks, designed to extract multiscale features efficiently. These blocks are interspersed with SE modules, enhancing the network's ability to focus on relevant spatial and channel-wise information. The compound scaling method is applied to optimize the network's depth, width, and input resolution, balancing computational cost and accuracy. Module 2 is introduced as the starting point for the first sub-block of all seven main blocks except the first one. Its steps involve a depthwise 2-D convolutional layer (Conv2D), batch normalization, activation, zero padding, and another depthwise Conv2D, followed by batch normalization and activation. Module 3 serves as a skip connection to all the sub-blocks, consisting of global average pooling, rescaling, a Conv2D layer, and a final Conv2D layer. Finally, the extracted features are passed through fully connected (FC) layers to generate the final output, demonstrating the streamlined and efficient design of EfficientNet-B7 for feature extraction tasks.

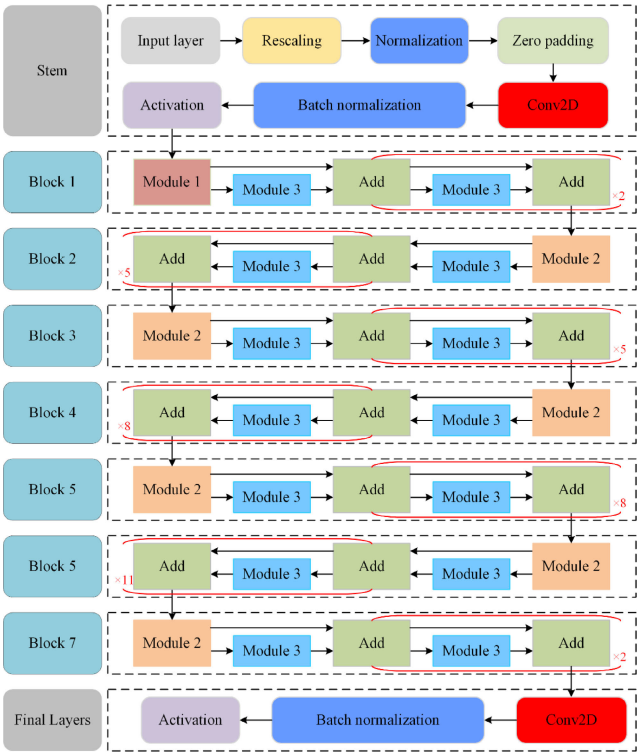


Fig. 2. Architectural workflow of EfficientNet-B7 with compound scaling strategy.

In the EfficientNet-B7, the input layer is first divided into fixed-size patches, which are then flattened into vectors. Each patch is transformed into an embedding using a linear projection [49]. This operation converts the inputs into a sequence of tokenized patches, which are suitable for processing in the transformer-like network architecture. To prepare the inputs for compatibility with subsequent layers of the model, they are transformed from a 2-D feature map into a 1-D token space. The expression for the embedded patch is

$$X_i = W \cdot I_{input} + b \tag{1}$$

where X_i represents the ith embedded patch, and W is the weight matrix, used to project the input patch to a higher dimensional space, I_{input} is the flattened representation of the ith input patch, and b is the bias term. This formula represents the transformation of the inputs into tokenized patches that can be processed by the subsequent layers in EfficientNet-B7. The learned weight matrix W and the bias b are crucial in embedding the patches into an appropriate high-dimensional

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space for further feature extraction. This embedding process enables the model to capture rich spatial and semantic information from the raw inputs, thereby enhancing its ability to learn complex feature representations in the later stages of the network. In this study, all input images are uniformly resized to a resolution of 128×128 to ensure consistency across the embedding process.

EfficientNet-B7 utilizes depthwise separable convolutions to reduce the number of parameters and computational costs while still maintaining the ability to extract rich features. The depthwise convolution applies a single filter per input, followed by a pointwise convolution to combine the output channels. To efficiently extract features from the inputs while maintaining computational efficiency, the convolution is split into spatial filtering and channel mixing. Its equation can be expressed as

$$\begin{aligned} F_{dw} &= Depthwise\,(Conv\,(X)) \\ F_{pw} &= Pointwise\,(Conv\,(F_{dw})) \end{aligned} \tag{2}$$

where F_{dw} is the feature map after depthwise convolution, F_{pw} is the feature map after pointwise convolution, and X is the input. The depthwise convolution allows EfficientNet-B7 to efficiently process feature maps while reducing the model complexity. It ensures that spatial relationships are preserved, and the pointwise convolution mixes channels to capture higher level features.

The SE block is designed to recalibrate the feature maps by applying a gating mechanism that dynamically adjusts the weights of the channels. This block learns which channels are important for a specific task and amplifies or suppresses them accordingly.

The compound scaling strategy is a key feature of EfficientNet-B7's design. It simultaneously scales the network's depth, width, and resolution in a balanced manner, using a set of scaling coefficients. This approach ensures proportional growth across all aspects of the network, optimizing both accuracy and computational efficiency. By fine-tuning the balance between depth, width, and resolution, this strategy enhances performance while minimizing computational costs. The corresponding equation can be expressed as

$$\begin{aligned} F_{ds} &= \alpha^{d} \\ F_{ws} &= \alpha^{w} \\ F_{rs} &= \alpha^{r} \end{aligned} \tag{3}$$

where α is the compound scaling coefficient, $d, w,$ and r denote the scaling factors for depth, width, and resolution, respectively, F_{ds} denotes the depth scaling factor, F_{ws} denotes the width scaling factor, and F_{rs} denotes the resolution scaling factor. The scaling coefficients $d, w,$ and r allow the model to grow in a balanced way, resulting in a better tradeoff between computational efficiency and accuracy.

The FC layer is the final step in the EfficientNet-B7. It takes the aggregated feature map and transforms it into the output space. The FC layer applies a linear transformation to map the high-dimensional feature vector to a task-specific output. The extracted features from the deep network are mapped into the final prediction space, allowing the model to effectively perform the designated task. The corresponding equation can be expressed as

$$y_{out_e} = W_{FC} \cdot F_{agg_eff} + B_{FC} \tag{4}$$

where y_{out_e} is the feature map output, W_{FC} is the weight matrix for the FC layer, F_{agg_eff} is the aggregated feature vector, and B_{FC} is the bias term for the FC layer. It captures the model's learned representation and generates the final output, making it crucial for the performance of the model.

The integration of depthwise separable convolutions, SE blocks, and compound scaling enables EfficientNet-B7 to efficiently capture hierarchical and multiscale features, delivering high performance on vision tasks. This design achieves superior accuracy while minimizing computational cost, making it highly suitable for large-scale image processing applications.

2) Pretrained PVT (Enhancing Global Context for Precise Remote Sensing Image Segmentation): PVT is a vision transformer that captures both local and global contextual information through a pyramid structure. Unlike traditional transformers with fixed-size patches, PVT processes images at multiple scales, enabling hierarchical feature representation across varying levels of detail [50]. In addition, PVT utilizes a multistage transformer block that progressively downsamples inputs, efficiently handling large-scale remote sensing imagery while preserving critical high-resolution details. This is particularly beneficial for landslide detection, enabling accurate segmentation of complex terrains and boundaries. By capturing long-range dependencies and contextual relationships, PVT maintains computational efficiency and fine-grained spatial details, making it highly effective for semantic segmentation and object recognition in landslide monitoring.

Fig. 3 illustrates the detailed workflow of PVT. The process begins with the inputs being divided into nonoverlapping patches, which are then embedded into a sequence of tokens. These tokens are passed through the initial stage of the transformer, where the model captures both local and global context through self-attention mechanisms. As the data progresses through multiple stages, the feature map is downsampled progressively, allowing the model to extract hierarchical features at different spatial resolutions. The pyramid structure ensures that the model captures both fine-grained details and high-level semantic information. At each stage, the feature map is refined through attention operations, allowing PVT to maintain rich spatial features while improving computational efficiency. Finally, the output from each stage is aggregated, producing a comprehensive feature representation suitable for segmentation.

In the initial step of the PVT, the inputs are divided into nonoverlapping patches, which are then flattened and embedded into a high-dimensional feature space. This process transforms the 2-D feature map into a 1-D sequence of tokens, suitable for processing by the transformer network. The inputs are transformed into a sequence of tokens, enabling compatibility with the transformer architecture.

The self-attention mechanism enables the model to capture both local and global contextual information by computing the relationships between all input patches. The mechanism computes attention scores based on the similarity between

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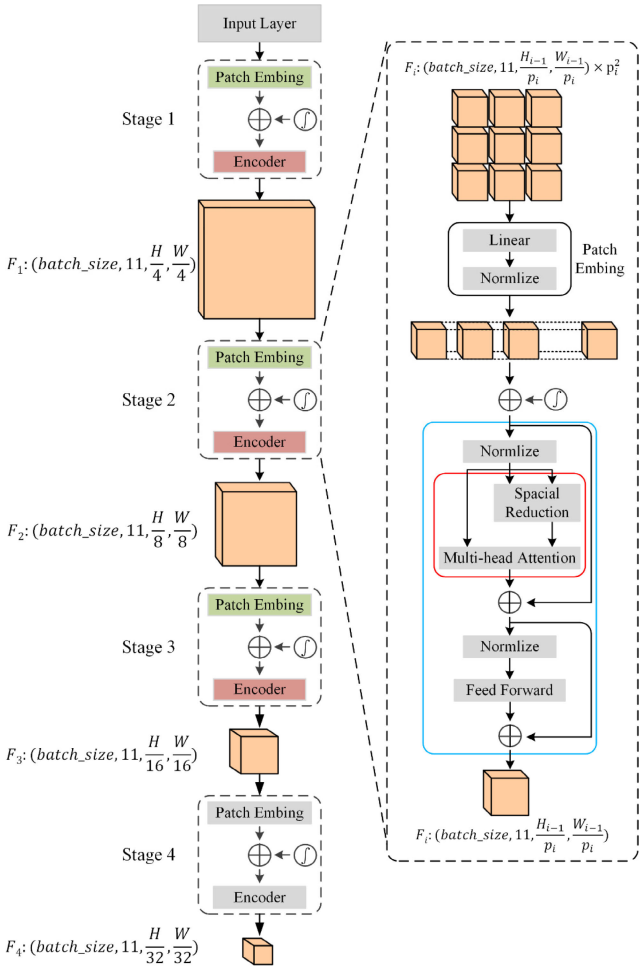


Fig. 3. Workflow of the PVT for multiscale feature extraction.

each token and all other tokens, allowing the model to focus on relevant features from across the entire feature map [1]. The model captures long-range dependencies between input patches, generating a rich and context-aware representation of the inputs. The corresponding equation can be expressed as

$$Attention\,(Q,K,V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

where Q is the query matrix, representing the current token's feature, K is the key matrix, representing all tokens' features, V is the value matrix, representing the feature values to be aggregated, d_k is the dimension of the key vectors, and $Softmax()$ is the softmax function. The attention mechanism calculates the weights between token pairs, enabling each token to focus on others in the image. This process captures both local and global relationships, enhancing the model's ability to generate contextually rich representations for further processing.

PVT employs a pyramid structure to progressively downsample the feature map. This multistage approach allows the model to capture hierarchical features at different spatial resolutions, enhancing the model's ability to detect both fine-grained details and high-level semantics. The spatial resolution is gradually reduced at each level, retaining essential semantic information while enhancing computational efficiency and preserving critical features. The corresponding equation can be expressed as

$$F_l = Downsample\,(F_{l-1}) \tag{6}$$

where F_l represents the feature map at the lth layer, and $Downsample()$ denotes the operation of reducing the feature map's spatial resolution. The pyramid structure progressively reduces the resolution of the feature map while retaining essential semantic information. This approach allows the model to efficiently process large-scale images, focusing on relevant patterns while minimizing computational cost.

After multiple stages of downsampling, the feature maps from each stage are aggregated. This aggregation step enables the model to combine features from different scales, creating a more comprehensive and hierarchical representation of the inputs. The features extracted at various scales are combined, allowing the model to integrate both fine-grained details and high-level semantic information for subsequent tasks. The corresponding equation is

$$F_{agg_pvt} = Concat\,(F_1, F_2, \ldots, F_n) \tag{7}$$

where F_{agg_pvt} is the aggregated feature map, $F_1, F_2, \ldots, F_n$ represent the feature maps from different pyramid levels, and $Concat()$ denotes the concatenation operation. The aggregation of features from different pyramid stages creates a rich, multiscale representation, ensuring that information from all levels is preserved and effectively utilized. This enhances the model's ability to make accurate predictions across diverse tasks.

After feature aggregation, the final feature map is passed through an FC layer to generate the output prediction. The FC layer maps the aggregated features into segmentation masks. Aggregated multiscale features are mapped to the output space, facilitating their use in downstream tasks. This process ensures that the extracted information is effectively utilized for the target application. The corresponding equation is

$$y_{out_p} = FC\left(F_{agg}\right) \tag{8}$$

where y_{out_p} is the feature map output, and $FC()$ refers to the FC layer, which maps the aggregated features to the output space. The final transformation of the aggregated features into segmentation masks leverages the extracted hierarchical information, enabling the model to make accurate predictions and complete the processing pipeline.

By utilizing these operations, PVT effectively extracts and aggregates multiscale features, balancing computational efficiency and performance. The pyramid structure combined with the self-attention mechanism allows for more efficient extraction of both global and local contexts, making PVT highly effective for a wide range of vision tasks.

3) UNet++ Encoder (Optimizing Hierarchical and Granular Feature Representation): The U-Net++ encoder is pivotal in efficiently extracting hierarchical features through nested skip connections, which link early and later layer feature maps for enhanced information reuse. This design enables multiscale feature extraction by progressively downsampling inputs while retaining essential spatial details [40]. Successive convolutions further refine feature representations, improving

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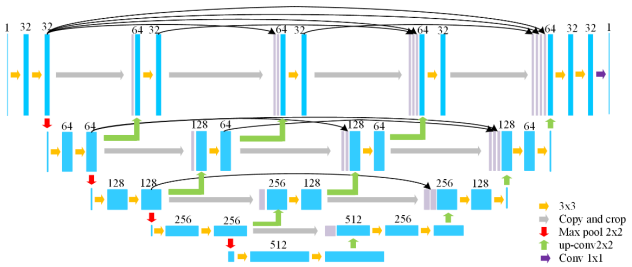


Fig. 4. Workflow of the U-Net++.

extraction accuracy. Its ability to capture complex features at various scales makes it particularly effective for tasks like landslide segmentation, where detailed image understanding and precise boundary delineation are crucial.

Fig. 4 illustrates the detailed workflow of the U-Net++ encoder, which processes the input through a series of convolutional layers, progressively extracting multiscale feature maps to capture hierarchical spatial information. The architecture employs nested skip connections, where the feature maps from lower layers are passed to higher layers to enhance contextual information and improve feature reuse.

The feature maps are then processed through downsampling operations to reduce spatial dimensions while preserving critical features. At each level, the convolutional layers refine the feature representations by applying further convolutions and activation functions, enhancing the ability to capture intricate patterns and structures in the inputs. Finally, the encoder outputs a set of enriched feature maps that contain both fine-grained details and high-level abstractions, ready to be passed to the decoder for reconstruction. This multiscale feature extraction and hierarchical information flow make the U-Net++ encoder highly effective for tasks that require detailed and context-aware feature extraction.

The U-Net++ encoder begins by extracting hierarchical features from the inputs using convolutional operations. This process transforms spatial information into high-level feature representations. Convolutional filters are applied to capture local patterns within the input image, generating feature maps at varying resolutions. This process enables the extraction of detailed spatial information across multiple scales. The corresponding mathematical expression is

$$F_l^0 = Conv2D\left(I_{input}, W_l^0\right) + b_l^0 \tag{9}$$

where F_l^0 is the feature map at layer l of the first block(0), W_l^0 is convolutional kernel weights at layer l of the first block(0), and b_l^0 is the BIAS term at layer l of the first block(0). This operation initiates hierarchical feature extraction through convolutional layers, allowing the model to capture spatial patterns effectively.

Nested skip connections ensure the encoder's feature maps are refined iteratively at different depths. Each level incorporates feature maps from prior levels for improved context understanding. This approach effectively combines lower resolution features with higher level abstractions, ensuring improved contextual understanding. The process can be mathematically represented as

$$F_l^k = Concat\left(F_{l-1}^{k-1}, F_{l-1}^k, F_l^{k-1}\right) \tag{10}$$

where F_l^k is feature map at layer l of first block(k), $Concat()$ is the concatenation function, F_{l-1}^{k-1}, F_{l-1}^k are the feature maps from previous layers contributing to the current map, and F_l^{k-1} is the feature map from the same layer but the preceding block. Nested connections enhance feature reuse and representation learning, providing richer information to subsequent layers.

Downsampling reduces the spatial dimensions of feature maps while retaining critical information, enabling the network to learn compact, high-level features. This process enhances computational efficiency and facilitates contextual understanding. The mathematical representation of downsampling can be expressed as

$$F_{l+1}^0 = MaxPool2D\left(F_l^k\right) \tag{11}$$

where F_{l+1}^0 is the downsampled feature map for the next layer, and $MaxPool2D()$ denotes the max pooling operation to reduce spatial dimensions. Downsampling balances computational cost with feature richness by progressively condensing spatial dimensions.

Successive convolutions further refine and enhance the feature maps by learning more complex patterns. To iteratively enhance feature representations at each level across the three encoders, ensuring progressively refined and context-aware features. The corresponding equation is expressed as

$$F_l^k = ReLU\left(Conv2D\left(F_l^{k-1}, W_l^k\right) + b_l^k\right) \tag{12}$$

where $ReLU()$ denotes the rectified linear unit activation function, W_l^k is the kernel weights at layer l of the first block(k), and b_l^k is the bias term at layer l of the first block(k). This step adds nonlinearity and enables the network to learn intricate feature representations, vital for downstream tasks.

Features extracted at different levels are passed to the decoder via nested skip connections, ensuring a seamless flow of information [2]. To propagate refined feature maps effectively to subsequent decoding stages, facilitating accurate reconstruction of spatial and semantic details. The corresponding equation is expressed as

$$F_{out_u} = \left\{F_l^k | \forall l, k\right\} \tag{13}$$

where F_{out_u} is the set of all feature maps to be passed to the decoder. The final feature aggregation consolidates information across layers, providing a rich feature set for the decoder to reconstruct high-resolution outputs.

The U-Net++ encoder boosts performance by capturing multiscale features through nested skip connections, enabling effective feature reuse and integration of fine-grained details with high-level abstractions. By downsampling and refining features at multiple resolutions, it captures diverse spatial patterns, enhancing robustness and accuracy for precise image segmentation in applications such as medical imaging, remote sensing, and object detection.

B. Feature Fusion

Feature fusion is a key component in enhancing the discriminative power of the model by combining multiscale feature

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maps extracted from various parts of the network. U-Net++ combines hierarchical feature maps from the encoder through feature fusion, improving multiscale representation. These fused features are then passed through convolution layers to reduce the dimensionality and generate a more compact and informative representation. The convolution operations are designed to aggregate the multiscale features into a unified feature map while preserving essential spatial information. This step is crucial for facilitating subsequent decoding and ensuring that the decoder has access to rich, multiscale information. Its equation can be expressed as

$$y^i_{fuse} = Conv2D\left(Concat\left(\left(F^i_{out_e}, F^i_{out_p}, F^i_{out_u}\right), W_{fuse}\right)\right) \tag{14}$$

where y^i_{fuse} is the fused output feature map at step i, and W_{fuse} is the weight matrix for the 1×1 convolution, which reduces the concatenated feature map to a fixed channel size for further processing in the decoder.

U-Net++ achieves feature fusion by concatenating multiscale feature maps from encoder blocks, blending spatial and semantic information. A 1×1 convolution is then applied to reduce dimensionality, compressing the fused feature maps while preserving critical information and recalibrating channel weights. This streamlined operation enhances computational efficiency and provides the decoder with rich, compact features, enabling improved accuracy and generalization, making it particularly effective for landslide detection in remote sensing imagery.

C. UNet++ Decoder

The decoder in the U-Net++ architecture, as shown in Fig. 4, plays a crucial role in reconstructing the input image by upsampling the feature maps and refining spatial resolution. Through a series of deconvolutions and concatenations, the decoder progressively enhances the feature maps from the encoder, improving the spatial details essential for accurate image interpretation [42]. This process ensures the integration of both low-level and high-level features, which is particularly vital for landslide detection in remote sensing imagery. The decoder's operations enhance model performance by refining multiscale information, allowing for precise delineation of landslide features and improved recognition accuracy.

Upsampling via transposed convolution increases the spatial resolution of feature maps while preserving key features, enhancing low-resolution details effectively [43]. Transposed convolutions are often used to perform this upsampling, which works by learning filters that increase the dimensions of the input feature map. Its equation can be expressed as

$$y_{up} = Conv2D\left(Transpose\left(y_{fuse}, W_{up}\right)\right) \tag{15}$$

where y_{up} is the upsampled output feature map, y_{fuse} is a fused output feature map, and W_{up} is the learned weight matrix used for upsampling. The upsampling step is crucial for restoring the original spatial resolution of the inputs, which is vital for tasks like landslide detection, where pixel-level accuracy is essential. This process ensures the recovery of fine-grained spatial details, enabling precise predictions and enhancing the model's ability to identify and delineate landslide features effectively.

Concatenation with skip connections merges feature maps from earlier layers of the encoders with the upsampled feature maps. This step allows the decoder to leverage low-level spatial information from earlier in the network while also considering the high-level semantic features from deeper layers. Its equation can be expressed as

$$y_{concat} = Concat\left(y_{up}, x_{skip}\right) \tag{16}$$

where y_{concat} is the concatenated feature map, and x_{skip} is the feature map from the corresponding encoder layer that provides low-level spatial details. The concatenation operation joins the upsampled feature map. y_{up} with the skip connection x_{skip} along the channel dimension. This allows the decoder to use both the high-level features from the deeper layers and the fine-grained spatial features from the earlier layers, combining their strengths. Concatenation with skip connections helps preserve fine-grained details in the output, which is crucial for high-resolution tasks like segmentation. It mitigates the problem of losing low-level spatial features as the network deepens, making the decoder more effective at reconstructing the original inputs.

After the feature maps are concatenated, convolution operations are applied to further refine them, enhancing the extracted features for more accurate landslide detection in remote sensing images and preparing them for the final prediction. The equation can be expressed as

$$y_{conv} = Conv2D\left(y_{concat}, W_{conv}\right) \tag{17}$$

where y_{conv} is the output feature map after convolution, and W_{conv} is the convolutional filter applied to the concatenated feature map. The convolution operation is performed on the concatenated feature maps to generate refined feature representations. This step allows the network to learn spatial patterns and relationships between the features, enabling it to produce more meaningful output. This convolutional step refines the feature map and helps it adapt to the specific task at hand. By processing the concatenated features, the decoder can generate high-quality representations that are crucial for achieving accurate results in tasks like segmentation.

The final step in the decoder generates the desired output, a landslide segmentation mask for remote sensing imagery. This is achieved through a final convolution that reduces the number of channels to match the output channels, ensuring accurate and precise landslide detection. Its equation can be expressed as

$$y_{out} = Conv2D\left(y_{conv}, W_{final}\right) \tag{18}$$

where y_{out} is the final output, and W_{final} is the final convolution filter, which reduces the number of channels to the desired output dimensions. The final convolution reduces the dimensionality of the feature map to match the desired output. The final convolution step is crucial for generating the desired output. It ensures that the feature map produced by the decoder is properly transformed into the format required for the segmentation task.

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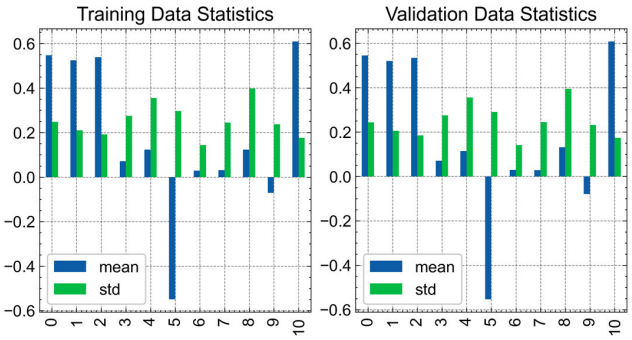


Fig. 5. Feature extraction for landslide monitoring: spectral, vegetation, and topographical indicators.

The U-Net++ decoder involves a series of steps that progressively refine and upsample the feature maps from the encoder. Starting with transposed convolution for upsampling, followed by concatenation with skip connections to incorporate low-level spatial details, and then applying convolutions for further refinement, the decoder prepares the outputs. The final convolution reduces the feature map's channel size to generate the output, specifically for landslide detection in remote sensing imagery. This step ensures the model accurately segments landslide areas, providing precise and actionable insights for monitoring and analysis.

III. Experiments and Results

This section presents experiments conducted on the Landslide4Sense dataset, which comprises three subsets: the development, evaluation, and test sets. Only the development set, which provides labels, is used in this study. It contains 3799 multispectral images acquired from Sentinel-2 and supplemented with ALOS PALSAR data. Each image consists of 14 bands, including 12 Sentinel-2 spectral bands, slope data, and elevation data, all resized to 128×128 pixels. The original spatial resolutions of the bands range from 10 to 60 m per pixel. Each image is paired with a binary ground truth mask of the same dimensions, where landslide and nonlandslide pixels are labeled as one and zero, respectively. Although 58% of the images contain landslide labels, landslide pixels account for only 2.3% of all pixels, with per-image proportions ranging from 0.0061% to 47.53%. This pronounced class imbalance poses a considerable challenge for segmentation tasks.

A. Data Preprocessing

The model training was completed within approximately 50 epochs, with each epoch taking around 80 s, demonstrating good training efficiency. A learning rate of 0.0002 was adopted to ensure stable convergence. The batch size and number of workers were both set to 4, and no data augmentation techniques were applied. All experiments were conducted on an NVIDIA GeForce RTX 4060 Ti GPU, which provided stable computational performance throughout the training process. Random flipping and random rotation were employed as data augmentation strategies to address skewed pixel distributions.

Fig. 5 illustrates the statistical distribution of normalized features for the training and validation datasets. The train_test_split function is employed to randomly divide the data, ensuring that the distributions of the two sets are similar. The left panel presents the training set statistics, which include 11 key indicators: RGB, NDVI, slope, elevation, NDMI, GNDVI, Band 10, brightness, and BSI. Among these, the RGB channels show comparable distributions, with means around 0.55 and variances near 0.2. NDVI has a mean of 0.05 and a variance of 0.23, slope has a mean of 0.1 and a variance of 0.32, and elevation has a mean of −0.57 and a variance of 0.3. NDMI and GNDVI exhibit means of 0.02 with variances of 0.14 and 0.22, respectively. Band 10 has a mean of 0.1 and a variance of 0.4, while brightness and BSI demonstrate means of −0.08 and 0.6, with variances of 0.21 and 0.18, respectively. In this study, 11 out of the 14 channels from the Landslide4Sense dataset were used as input, with all channels initialized with equal weights at the start of training.

B. Feature Extraction and Model Construction

This section outlines the methodologies for feature extraction and model construction, detailing how critical landslide features are captured and processed using state-of-the-art architectures to achieve accurate monitoring through advanced feature representation and fusion techniques.

Fig. 6 illustrates the feature map outputs of the feature extraction layers for the three models, using the 77th image from the dataset as an example. The top row showcases the output from the EP-UNet++ feature extraction layer, which demonstrates a rich representation that balances both global and local information effectively. The middle row represents the output from the Efficient-UNet++ feature extraction layer, which places a stronger emphasis on capturing local details. The bottom row shows the output from the PVT-UNet++ feature extraction layer, which primarily focuses on global structural information. Overall, the visualization highlights the unique strengths of each model's feature extraction capability, with EP-UNet++ offering a more comprehensive and balanced representation.

Fig. 7 presents the ablation study results, showing the prediction outputs of EP-UNet++, PVT-UNet++, and Efficient-UNet++ on images 30–38 from the dataset. In image 30, EP-UNet++ incorrectly predicts a small area in the top-left corner as a landslide region, influenced by the feature extraction characteristics of Efficient-UNet++, while PVT-UNet++ achieves the most accurate prediction. For image 31, EP-UNet++ performs best, followed by PVT-UNet++, with Efficient-UNet++ performing worst. Image 32 predictions by EP-UNet++ and PVT-UNet++ are similar, with PVT-UNet++ slightly lagging. Images 33 and 35 contain no landslide regions, and all three models make completely correct predictions. In image 34, predictions are similar across the models, but all incorrectly classify a small area in the top-left corner. For image 36, EP-UNet++ performs best, followed by PVT-UNet++, with PVT-UNet++ yielding the least accurate result. In image 37, EP-UNet++ again performs best, while the other two models fail to predict any landslide regions. Overall, EP-UNet++ delivers high predictive accuracy and robustness, even in complex scenarios, reinforcing its effectiveness in precise

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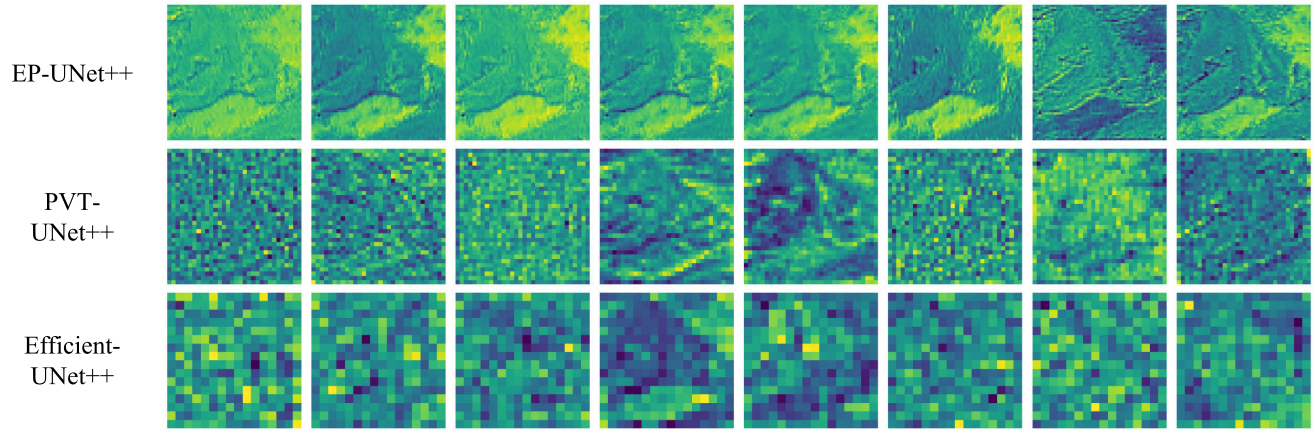


Fig. 6. Feature map visualization of the extraction layers across models.

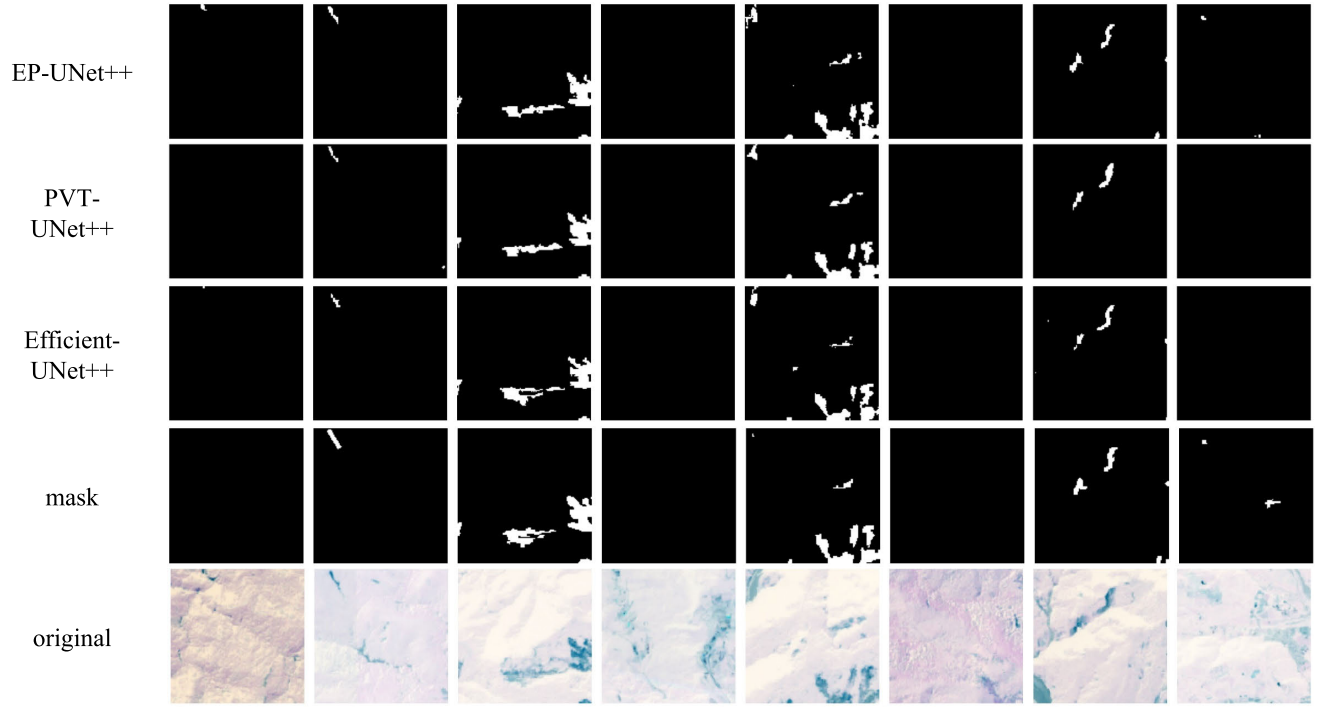


Fig. 7. Ablation study: prediction results on selected images (30–38).

landslide region identification and supporting reliable landslide monitoring.

C. Model Evaluation

The model performance is evaluated through an ablation study, comparing EP-UNet++ with PVT-UNet++ and Efficient-UNet++ in landslide detection. The evaluation includes receiver operating characteristic (ROC) curves and confusion matrices. These analyses assess the models' sensitivity, precision, and overall classification accuracy, highlighting the superior performance of EP-UNet++ in landslide detection.

Fig. 8 presents the ROC curve comparing the performance of three untrained models. EP-UNet++ demonstrates the most rapid improvement, achieving the highest area under the curve (AUC) value of 0.7, indicating its strong ability to distinguish between positive and negative samples even before training. PVT-UNet++ shows a competitive, albeit slightly lower, AUC of 0.68, reflecting moderate early stage performance. In contrast, Efficient-UNet++ exhibits the lowest

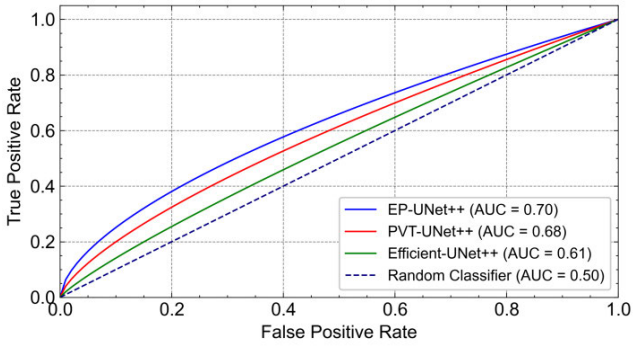


Fig. 8. True positive rate versus false positive rate for untrained models.

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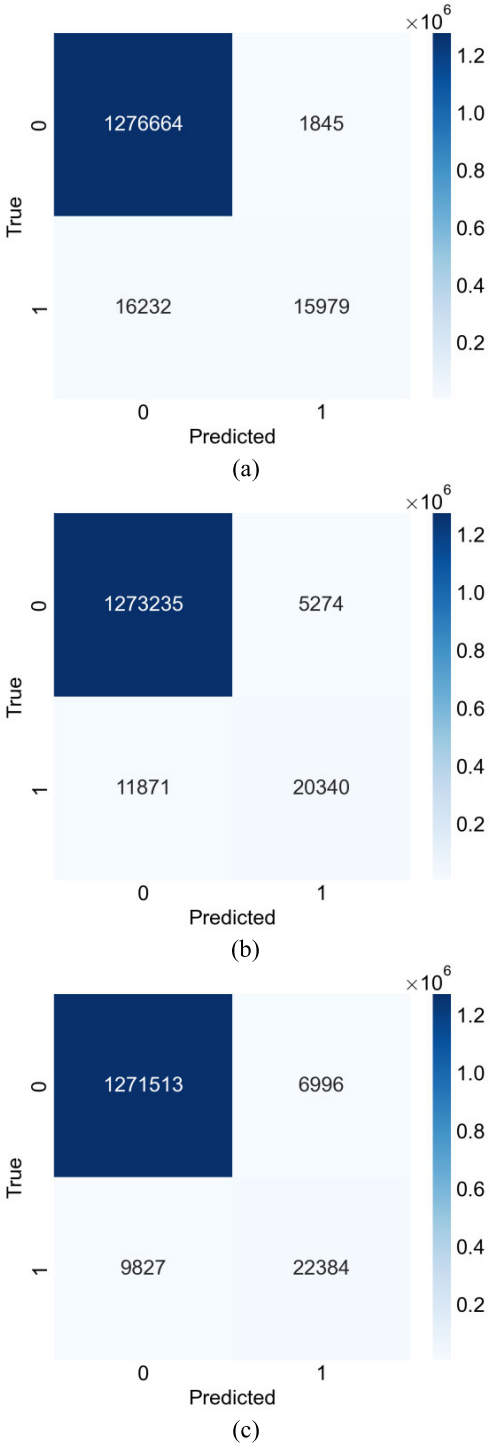


Fig. 9. Confusion matrices comparing the classification performance of (a) EP-UNet++, (b) PVT-UNet++, and (c) Efficient-UNet++.

AUC of 0.61, suggesting limited predictive capability at this stage. These results indicate that EP-UNet++ possesses a more advantageous initial feature representation, which may contribute to faster convergence during training. The clear performance gap at this early stage also suggests its greater potential for achieving superior results in the fully trained state.

Fig. 9 presents the confusion matrices for the three models, providing a detailed comparison of their classification performance in landslide detection. Fig. 9(a) illustrates EP-UNet++, which achieves 1 276 664 true negatives (0, 0), 1845 false negatives (1, 0), 16 232 false positives (0, 1), and 15 979 true positives (1, 1). Fig. 9(b) shows PVT-UNet++, yielding 1 273 235 true negatives, 5274 false negatives, 11 871 false positives, and 20 340 true positives. Fig. 9(c) depicts Efficient-UNet++, with 1 271 513 true negatives, 6996 false negatives, 9827 false positives, and 22 384 true positives. Overall, the results demonstrate that EP-UNet++ achieves the lowest false negative count, indicating superior sensitivity to landslide regions, while maintaining competitive true positive and false positive rates. This balance between sensitivity and precision highlights its robustness and effectiveness for accurate landslide region identification in challenging scenarios.

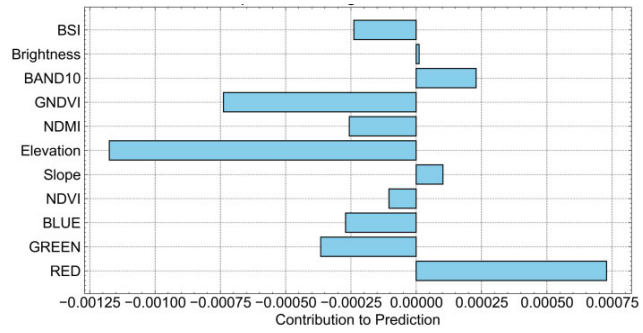


Fig. 10. LIME analysis of input features to landslide predictions.

Fig. 10 presents the local interpretable model-agnostic explanation (LIME) analysis of different input features in the model, illustrating their respective contributions to the prediction outcomes. Among the positive contributors, RED has the highest impact at approximately 0.00075, followed by Band 10 at around 0.00024, slope at 0.0001, and brightness at 0.00002. In terms of negative contributions, elevation shows the largest effect at approximately −0.0011, followed by GNDVI at −0.00075, GREEN at −0.0004, Blue at −0.00027, NDMI at −0.00025, BSI at −0.00023, and NDVI at −0.0001. Overall, this analysis highlights that spectral bands such as RED and Band 10 predominantly drive positive predictions, while topographical and vegetation indices, particularly elevation and GNDVI, exert a significant negative influence, providing insight into the relative importance of features in the model's decision-making process.

D. Comparison With Alternative Methods

To thoroughly assess the performance of the proposed EP-UNet++, this section presents a detailed comparison with several state-of-the-art models, including UNet [35], Swin-UNet [34], Mask2Former [52], DeeplabV3+ [53], and SegFormer [54]. It plays a critical role in benchmarking the proposed model, highlighting its strengths and weaknesses in comparison to leading alternatives.

Fig. 11 presents the comparative training performance of EP-UNet++ and several baseline methods across four key evaluation metrics. In Fig. 11(a), illustrating intersection over union (IoU), most baseline models start near 0.1 and gradually converge around 0.6, whereas EP-UNet++ begins at

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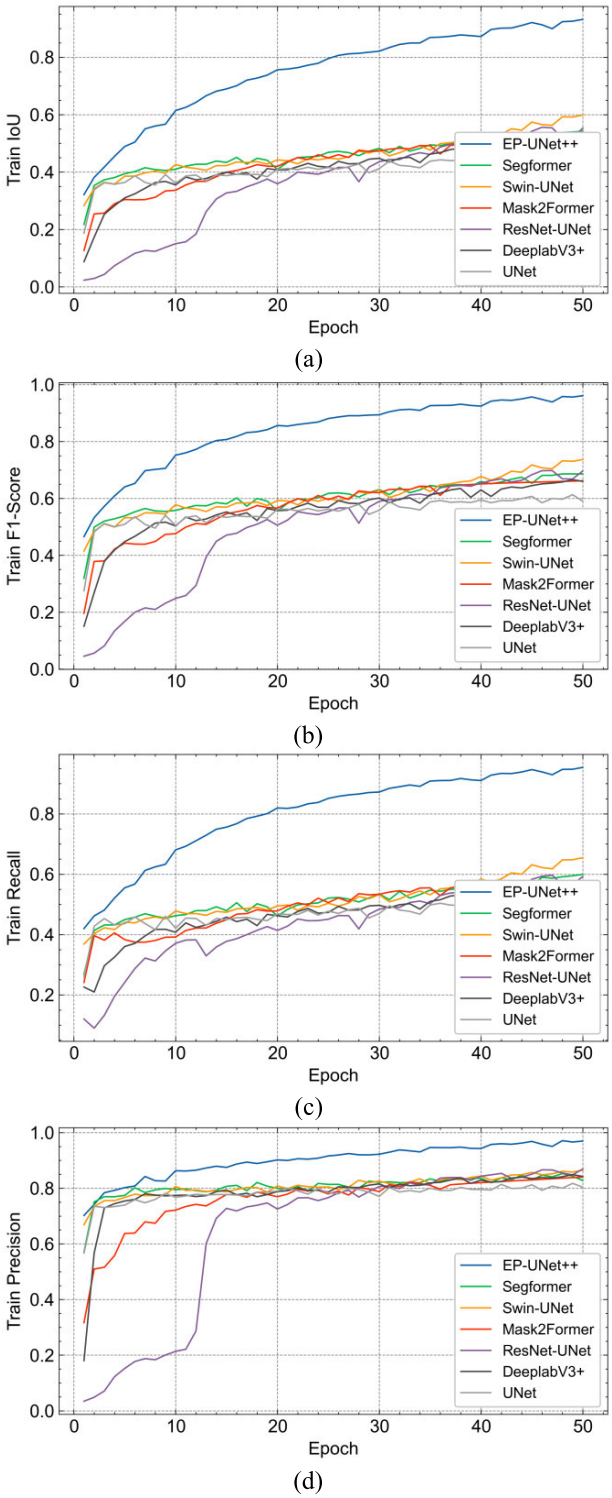


Fig. 11. (a) Train IoU, (b) $F1$-score, (c) recall, and (d) precision of EP-UNet++ and alternative models.

approximately 0.3 and rapidly progresses to a final value of 0.93, representing an improvement of roughly 55% over the best baseline and highlighting its faster learning and superior feature representation. Fig. 11(b) depicts the $F1$-score, where EP-UNet++ steadily increases from 0.45 to 0.96, significantly outperforming UNet, which rises only from 0.3 to 0.6, and other models that converge near 0.7, corresponding to an improvement of approximately 37%–57% compared to the baselines. For recall, shown in Fig. 11(c), EP-UNet++ advances from 0.45 to 0.95, whereas other models start around 0.1 and reach only 0.5, indicating a relative enhancement of 90% in sensitivity for detecting relevant features. In Fig. 11(d), showing precision, EP-UNet++ starts near 0.7 and converges to 0.97, compared to UNet with a modest improvement from 0.7 to 0.8 and other models stabilizing near 0.85, yielding a relative gain of approximately 14%–20%. Collectively, these results demonstrate that EP-UNet++ not only achieves higher final performance across all metrics but also exhibits faster convergence, reflecting its enhanced capability to effectively integrate local and global features for accurate and reliable landslide remote sensing image segmentation, with the four training results further confirming that EP-UNet++ shows no signs of overfitting.

Fig. 12 presents the prediction results of different models for images 10–17 from the dataset, illustrating their performance in landslide boundary detection and overall prediction accuracy. In the tenth image, EP-UNet++ achieves the best results, although a few landslide pixels in the central region are missed. SegFormer performs slightly worse, with several nonlandslide pixels in the upper right and central regions misclassified as landslide. Swin-UNet shows a similar trend but with more misclassified pixels. DeepLabv3++ performs slightly lower, missing multiple landslide pixels in the upper right and central areas. UNet exhibits similar results to DeepLabv3++, while ResNet-UNet underperforms in the upper right and fails to detect central landslide pixels. Mask2Former performs the poorest, barely identifying any landslide pixels. In the 11th image, which contains no landslide areas, all models except ResNet-UNet make correct predictions. For the 12th image, EP-UNet++, SegFormer, Swin-UNet, and UNet closely match the ground truth mask, whereas DeepLabv3++ and ResNet-UNet perform slightly worse, and Mask2Former again performs the worst.

For the 13th image, SegFormer achieves the best results but misclassifies multiple nonlandslide pixels in the central region, with EP-UNet++ slightly behind, missing some central landslide pixels. Other models show similar patterns, performing well in the lower left area but poorly in the central region. In the 14th image, EP-UNet++, Swin-UNet, DeepLabv3++, and UNet demonstrate good agreement with the ground truth, followed by SegFormer and ResNet-UNet, while Mask2Former fails to detect landslide pixels. In the 15th image, EP-UNet++ and SegFormer perform the best, although SegFormer misclassifies several nonlandslide areas, while Swin-UNet, DeepLabv3++, ResNet-UNet, and UNet show moderate performance with missed landslide pixels in the central region. Mask2Former performs the worst, barely detecting landslide pixels on the right. In the 16th image, EP-UNet++ clearly outperforms other models, which show similar but lower performance. Finally, in the 17th image, EP-UNet++ and Swin-UNet achieve the best detection, with other models performing comparably but less accurately. Overall, EP-UNet++ consistently demonstrates superior accuracy and boundary delineation across varied images, highlighting its

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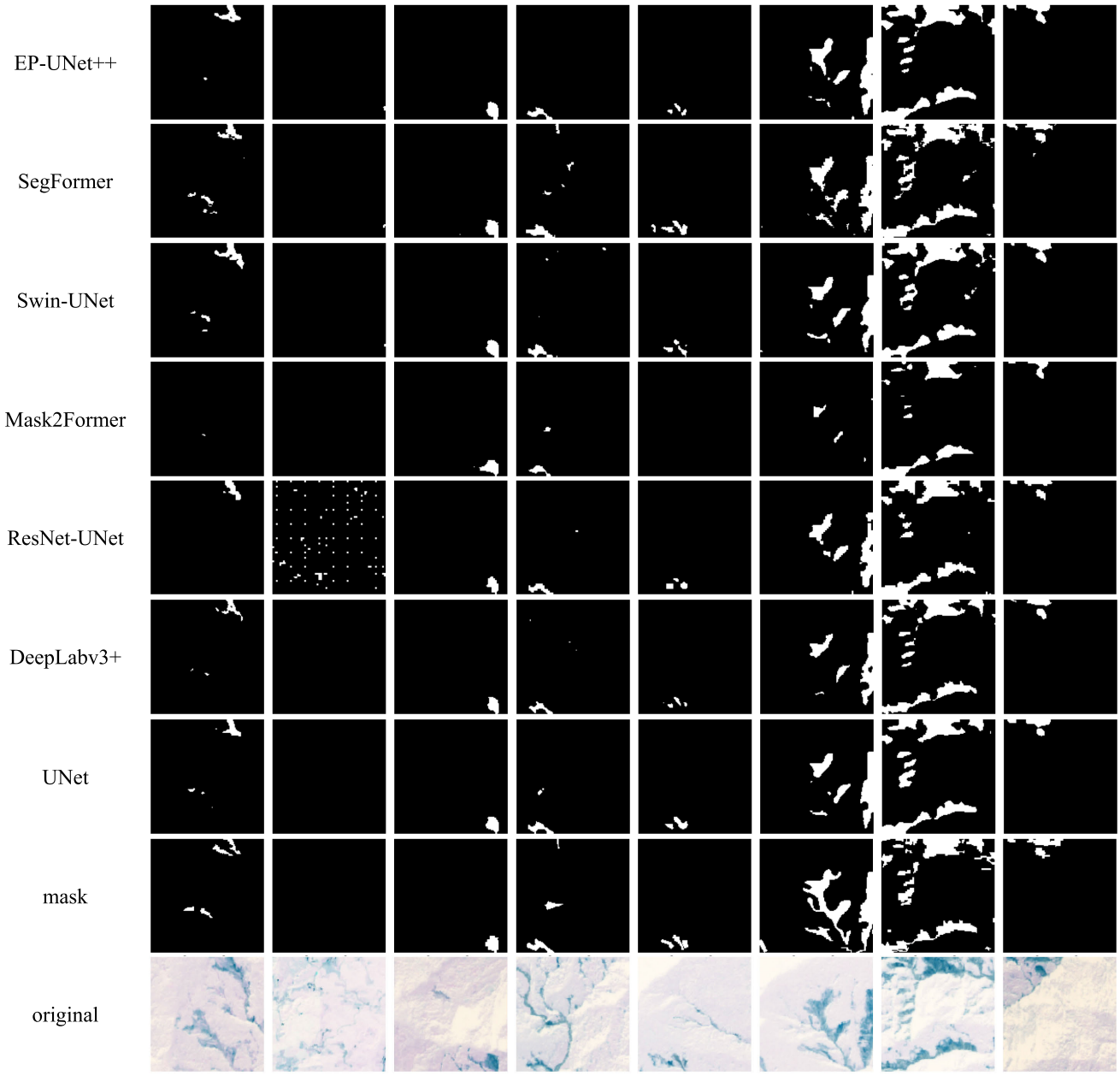


Fig. 12. Prediction results of different models on selected dataset images (10)–(18).

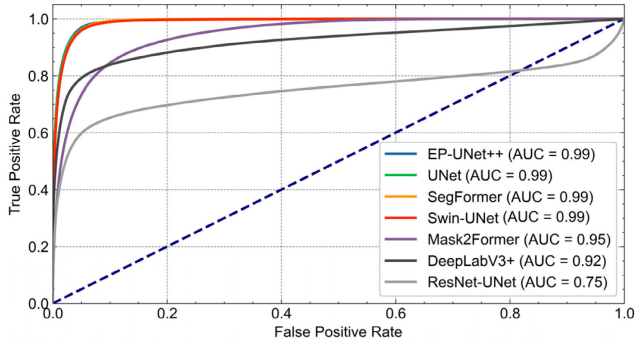


Fig. 13. ROC curve diagram for prediction results across trained models.

robustness and reliability for landslide detection in complex scenarios.

Fig. 13 presents the ROC curves of EP-UNet++ compared with several baseline models. EP-UNet++, SegFormer, Swin-UNet, and UNet all achieve an AUC of 0.99, with EP-UNet++ converging the fastest, followed by Swin-UNet, while SegFormer and UNet show similar convergence behavior. Mask2Former achieves an AUC of 0.97, and DeepLabv3+ reaches 0.94, initially converging more quickly than Mask2Former. The lowest performance is observed for ResNet-UNet, with an AUC of 0.82. These results indicate that EP-UNet++ achieves the highest predictive accuracy and exhibits the fastest convergence among the evaluated models.

Table I presents a comprehensive comparison of multiple models across five evaluation metrics: mean intersection over union (mIoU), mean dice coefficient (mDice), specificity, FLOPs, and inference time. Among the baseline models, Mask2Former exhibits the lowest segmentation performance,

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS IN LANDSLIDE DETECTION

Model	mIoU	mDice	mSpecificity	FLOPs/G	Time/ms
SegFormer	69.51	82.01	0.971	0.44	5.87
Swin-UNet	71.25	83.21	0.969	2.94	8.43
Mask2Former	68.43	81.26	0.971	2.83	20.54
ResNet-UNet	68.17	81.07	0.966	3.21	11.40
DeepLabv3+	67.83	80.83	0.969	3.10	5.05
UNet	70.24	82.52	0.0962	20.46	3.99
EP-UNet++	74.15	85.16	0.979	52.62	38.23

achieving an mIoU of 68.43 and an mDice of 81.26, despite maintaining a relatively high specificity of 0.971. ResNet-UNet and UNet demonstrate moderate performance, with mIoU values of 68.17 and 70.24 and mDice values of 81.07 and 82.52, respectively, although UNet shows an unusually low specificity of 0.0962. DeepLabv3+ and SegFormer achieve slightly higher mIoU and mDice scores, with SegFormer reaching 69.51 and 82.01, and DeepLabv3+ attaining 67.83 and 80.83, both maintaining specificity around 0.97. Swin-UNet emerges as the second-best model, achieving an mIoU of 71.25 and an mDice of 83.21, with a specificity of 0.969. EP-UNet++ outperforms all competitors across every metric, achieving the highest mIoU of 74.15, mDice of 85.16, and specificity of 0.979. Compared to Swin-UNet, EP-UNet++ shows improvements of approximately 4.0% in mIoU and 2.3% in mDice, along with notable gains in specificity. In terms of efficiency, SegFormer and DeepLabv3+ have the lowest computational cost with FLOPs of 0.44 G and 3.10 G and inference times of 5.87 and 5.05 ms, respectively, whereas EP-UNet++ requires 52.62 G FLOPs and 38.23 ms but delivers the best accuracy–efficiency tradeoff. Overall, EP-UNet++ demonstrates clear advantages in segmentation accuracy, robustness, and practical applicability, highlighting its suitability for precise landslide remote sensing image segmentation.

E. Application on Enshi Landslide Imagery

To further evaluate the practical applicability of the proposed EP-UNet++, a real-world landslide case in Enshi, Hubei Province, China, was selected for validation. The satellite image, captured by Sentinel-2 on July 21, 2020, depicts the Ma Zhe Village landslide along the left bank of the Qingjiang River in Tunpu Township. Triggered by continuous heavy rainfall, the landslide body at Shazi Dam advanced into the river, temporarily blocking the Qingjiang River and forming a barrier lake. This case with complex terrain and active deformation tests the model's ability to accurately identify landslides and their extents.

Fig. 14 illustrates the feature extraction process for landslide monitoring, encompassing RGB, NDVI, slope, elevation, GNDVI, NDMI, brightness, BSI, and Band 10. Each feature captures critical environmental and spectral information essential for analyzing terrain and vegetation dynamics. RGB channels represent visible spectral data for identifying surface characteristics, while NDVI and GNDVI highlight vegetation

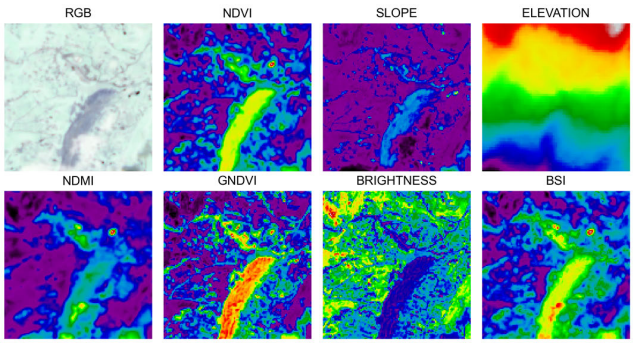


Fig. 14. Feature extraction for Enshi landslide monitoring: spectral, vegetation, and topographical indicators.

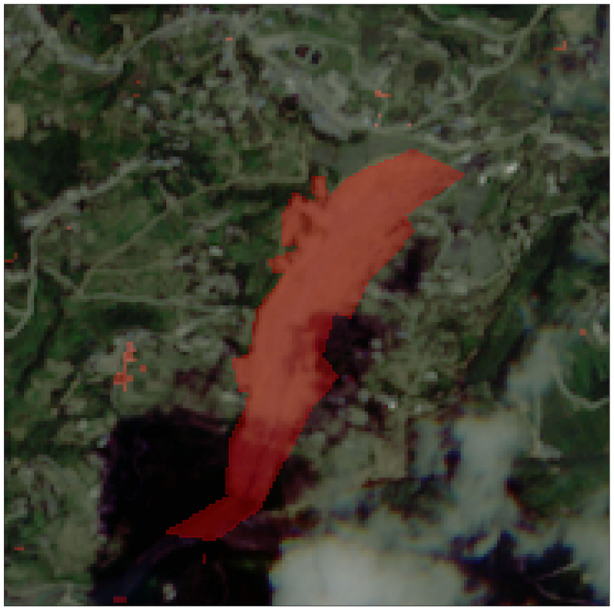


Fig. 15. EP-UNet++ prediction results overlaid on Sentinel-2 Image of the Enshi landslide.

health and chlorophyll content, indicating susceptibility to landslides. Slope and elevation provide topographical insights, crucial for understanding water flow and erosion processes. Brightness measures surface reflectance, aiding in identifying land cover changes, and BSI distinguishes bare soil from vegetation, vital for monitoring sparsely vegetated areas. Band 10 contributes thermal and soil moisture analysis for assessing soil stability. Notably, in the eight feature maps, the landslide regions exhibit distinct patterns and contrasts compared to the surrounding areas, clearly differentiating affected zones from stable terrain. Together, these features enable a comprehensive and precise framework for landslide susceptibility analysis, facilitating robust terrain monitoring.

Fig. 15 presents the prediction results of the EP-UNet++ overlaid on the Sentinel-2 satellite image of the Enshi landslide. The figure demonstrates that the model accurately delineates the landslide region, with predicted boundaries closely matching the actual affected area. Despite the presence of clouds in the original remote sensing image, the model's predictions remain largely unaffected, highlighting its robustness to atmospheric interference. Overall, these results confirm the effectiveness of EP-UNet++ in real-world

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landslide detection, demonstrating its capability to provide precise and reliable mapping even under challenging imaging conditions.

IV. Discussion

The results demonstrate the superior performance of EP-UNet++ in landslide detection compared to models including UNet, SegFormer, Swin-UNet, DeepLabv3++, ResNet-UNet, and Mask2Former. EP-UNet++ consistently achieves the highest scores across all metrics, including mIoU, mDice, specificity, and accuracy, thanks to its integration of PVTs with advanced encoder–decoder structures. This enables the model to capture both local and global contextual information, facilitating precise segmentation in complex terrains. Feature analysis further confirms its ability to effectively leverage spectral, topographical, and vegetation information, even in the presence of clouds.

EP-UNet++ outperforms other models in both accuracy and boundary delineation. Its mIoU and mDice are approximately 4% higher than the next best model, Swin-UNet. Qualitative results on dataset images 10–17 show that EP-UNet++ accurately identifies landslide regions, whereas models such as Mask2Former, ResNet-UNet, and DeepLabv3++ struggle with fine-grained details or misclassify certain areas. Traditional convolution-based models like FCN and SegNet perform poorly due to limited multiscale contextual understanding.

Despite the clear advantages of EP-UNet++, some challenges remain. Certain images demonstrate missed pixels in central regions or slight misclassifications in complex areas, indicating that extremely heterogeneous terrains or ambiguous boundaries still pose difficulties. Comparatively, models such as UNet, ResNet-UNet, and Mask2Former underperform due to their limited capacity to capture multiscale contextual information, reflecting the necessity of combining transformer-based global feature extraction with convolutional local feature processing.

Although EP-UNet++ exhibits higher FLOPs and longer inference time than lighter models such as SegFormer, the computational overhead is justified by the substantial gains in segmentation accuracy and robustness, particularly when processing multichannel landslide imagery. The observed inference time of 38.23 ms per frame remains practical for high-resolution remote sensing applications, and the performance improvements provide a favorable tradeoff for real-world environmental monitoring, disaster prevention, and geospatial hazard assessment scenarios.

Overall, these findings underscore the significance of hybrid architectures for robust landslide monitoring. The integration of PVTs with advanced encoder–decoder structures allows EP-UNet++ to achieve precise, reliable, and resilient segmentation across varied scenarios. This comparative analysis validates the model's effectiveness and demonstrates its potential for real-world applications in environmental monitoring, disaster prevention, and geospatial hazard assessment.

V. Conclusion

This proposes the EP-UNet++ for landslide detection, combining EfficientNet, PVT, and advanced encoder–decoder architectures. The workflow includes preprocessing remote sensing data, extracting multiscale spectral and topographical features, constructing the model, and evaluating its performance using quantitative and qualitative metrics. Results show that EP-UNet++ outperforms models such as SegFormer, Swin-UNet, ResNet-UNet, DeepLabv3++, UNet, and Mask2Former in both accuracy and boundary delineation, particularly in complex terrain and under cloud interference. The study also examines the architectural strengths and weaknesses of each model, providing insights into performance disparities.

EP-UNet++ achieved the highest mIoU, mDice, specificity, and accuracy scores among all tested models. Its superior performance can be attributed to its ability to effectively capture both local and global contextual information, leveraging multiscale feature extraction and attention mechanisms. In contrast, convolution-based or simpler hybrid models exhibited lower segmentation performance, particularly in fine-grained boundary detection and heterogeneous terrain regions, highlighting the advantages of combining transformer-based global feature processing with convolutional local feature extraction.

The success of the proposed model underscores its potential for improving landslide detection and other remote sensing applications. Its ability to efficiently process multiscale features offers significant advantages for environmental monitoring, suggesting a new direction for combining transformer-based models with convolutional architectures in geospatial tasks. This approach could inspire future research in similar domains. while also providing substantial economic benefits through cost-efficient monitoring and the minimization of disaster-induced economic losses.

The proposed model demonstrates strong adaptability for newly occurring landslides, as its geospatial fusion strategy enhances generalization across diverse terrains. By leveraging multisource features beyond single-scene characteristics, the model effectively identifies landslides in previously unseen regions. This highlights its potential applicability in real-world scenarios where rapid and accurate detection of new landslides is essential for effective disaster response and risk management.

Future research should focus on expanding the model's applicability to diverse datasets, improving computational efficiency for real-time use, and incorporating additional environmental factors like soil moisture and vegetation type. Further studies could also explore integrating EP-UNet++ with other sensor data to enhance prediction accuracy and robustness in different geospatial contexts.

Acknowledgment

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REFERENCES

[1] B. Zhao, L. Su, Q. Xu, W. Li, C. Xu, and Y. Wang, "A review of recent earthquake-induced landslides on the Tibetan Plateau," *Earth-Sci. Rev.*, vol. 244, Sep. 2023, Art. no. 104534.

[2] M. Dai, D. Cui, Q. Chen, J. Wei, J. Wang, and G. Zhang, "Reactivated mechanism of a slow-moving landslide with two shear zones based on ring shear test and in situ monitoring," *Landslides*, vol. 21, no. 11, pp. 2617–2634, Nov. 2024.

[3] N. Casagli, E. Intrieri, V. Tofani, G. Gigli, and F. Raspini, "Landslide detection, monitoring and prediction with remote-sensing techniques," *Nature Rev. Earth Environ.*, vol. 4, no. 1, pp. 51–64, Jan. 2023.

[4] H. Thirugnanam, S. Uhlemann, R. Reghunadh, M. V. Ramesh, and V. P. Rangan, "Review of landslide monitoring techniques with IoT integration opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5317–5338, 2022.

[5] F. Xie, E. Larose, Q. Wang, and Y. Zhang, "In-situ monitoring of rock slope destabilization with ambient seismic noise interferometry in Southwest China," *Eng. Geol.*, vol. 312, Jan. 2023, Art. no. 106922.

[6] M. Su et al., "Combining geophysical methods, drilling, and monitoring techniques to investigate carbonaceous shale landslides along a railway line: A case study on Jiheng railway, China," *Bull. Eng. Geol. Environ.*, vol. 80, no. 10, pp. 7493–7506, Oct. 2021.

[7] Y. Hussain et al., "Review on the geophysical and UAV-based methods applied to landslides," *Remote Sens.*, vol. 14, no. 18, p. 4564, Sep. 2022.

[8] J. Boyd et al., "A linked geomorphological and geophysical modelling methodology applied to an active landslide," *Landslides*, vol. 18, no. 8, pp. 2689–2704, Aug. 2021.

[9] P. Lv, L. Ma, Q. Li, and F. Du, "ShapeFormer: A shape-enhanced vision transformer model for optical remote sensing image landslide detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2681–2689, 2023.

[10] X. Zhang, W. Yu, M.-O. Pun, and W. Shi, "Cross-domain landslide mapping from large-scale remote sensing images using prototype-guided domain-aware progressive representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 1–17, Mar. 2023.

[11] Z. Lu et al., "An iterative classification and semantic segmentation network for old landslide detection using high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4408813.

[12] G. Huang, S. Du, and D. Wang, "GNSS techniques for real-time monitoring of landslides: A review," *Satell. Navigat.*, vol. 4, no. 1, p. 5, Dec. 2023.

[13] N. H. Do et al., "Mechanism of a rainfall-induced landslide in a large-scale flume experiment on a weathered granite sand," *Geoenvironmental Disasters*, vol. 11, no. 1, p. 18, Jul. 2024.

[14] R. S. Tandon, V. Gupta, B. Venkateshwarlu, and P. Joshi, "An assessment of Dungale landslide using remotely piloted aircraft system (RPAS), ground penetration radar (GPR), and slide & RS2 softwares," *Natural Hazards*, vol. 113, no. 2, pp. 1017–1042, Sep. 2022.

[15] A. Manconi and A. C. Mondini, "Landslides caught on seismic networks and satellite radars," *Natural Hazards Earth Syst. Sci.*, vol. 22, no. 5, pp. 1–13, May 2022.

[16] D. B. Ichsandya et al., "Landslide assessment using interferometric synthetic aperture radar in Pacitan, East Java," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 12, no. 3, pp. 2614–2625, Jun. 2022.

[17] R. Tomás et al., "Advances on the investigation of landslides by space-borne synthetic aperture radar interferometry," *Geo-Spatial Inf. Sci.*, vol. 27, no. 3, pp. 602–623, May 2024.

[18] F. Catani, "Landslide detection by deep learning of non-nadiral and crowdsourced optical images," *Landslides*, vol. 18, no. 3, pp. 1025–1044, Mar. 2021.

[19] C. Ding, G. Feng, M. Liao, P. Tao, L. Zhang, and Q. Xu, "Displacement history and potential triggering factors of baige landslides, China revealed by optical imagery time series," *Remote Sens. Environ.*, vol. 254, Mar. 2021, Art. no. 112253.

[20] J. Cai et al., "Detection and characterization of slow-moving landslides in the 2017 Jiuzhaigou earthquake area by combining satellite SAR observations and airborne LiDAR DSM," *Eng. Geol.*, vol. 305, Aug. 2022, Art. no. 106730.

[21] M. Syzdykbayev, B. Karimi, and H. A. Karimi, "Persistent homology on LiDAR data to detect landslides," *Remote Sens. Environ.*, vol. 246, Sep. 2020, Art. no. 111816.

[22] L. Zhu, G. Wang, F. Huang, Y. Li, W. Chen, and H. Hong, "Landslide susceptibility prediction using sparse feature extraction and machine learning models based on GIS and remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] A. C. Mondini, "SAR amplitude exploitation for systematic landslide failure detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 1800–1803.

[24] C. Niu, H. Zhang, W. Liu, R. Li, and T. Hu, "Using a fully polarimetric SAR to detect landslide in complex surroundings: Case study of 2015 Shenzhen landslide," *ISPRS J. Photogramm. Remote Sens.*, vol. 174, pp. 56–67, Apr. 2021.

[25] L. Nava, O. Monserrat, and F. Catani, "Improving landslide detection on SAR data through deep learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] V.-H. Dang, T. B. Dieu, X.-L. Tran, and N.-D. Hoang, "Enhancing the accuracy of rainfall-induced landslide prediction along mountain roads with a GIS-based random forest classifier," *Bull. Eng. Geol. Environ.*, vol. 78, no. 4, pp. 2835–2849, Jun. 2019.

[27] W. Shi, M. Zhang, H. Ke, X. Fang, Z. Zhan, and S. Chen, "Landslide recognition by deep convolutional neural network and change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4654–4672, Jun. 2021.

[28] M. Gao, F. Chen, L. Wang, H. Zhao, and B. Yu, "Swin transformer-based multiscale attention model for landslide extraction from large-scale area," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4415314.

[29] Q. Zhang et al., "A graph-transformer method for landslide susceptibility mapping," *IEEE J. Select. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 14556–14574, 2024.

[30] H. Alhichri, A. S. Alswayed, Y. Bazi, N. Ammour, and N. A. Alajlan, "Classification of remote sensing images using EfficientNet-B3 CNN model with attention," *IEEE Access*, vol. 9, pp. 14078–14094, 2021.

[31] C. Huang, W. Wang, X. Zhang, S.-H. Wang, and Y.-D. Zhang, "Tuberculosis diagnosis using deep transferred EfficientNet," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 20, no. 5, pp. 2639–2646, Sep. 2023.

[32] S.-E. Hsieh et al., "7.6 a 70.85–86.27TOPS/W PVT-insensitive 8b word-wise ACIM with post-processing relaxation," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2023, pp. 136–138.

[33] Z. Xiao et al., "Predicting urban region heat via learning arrive-stay-leave behaviors of private cars," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 10843–10856, Oct. 2023.

[34] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[35] P. Manasi, G. Rojasree, G. Jahnavi, U. Yojitha, A. Saha, and T. Pal, "U-Net deep neural network-based landslide detection method," in *Proc. Int. Conf. Intell. Syst. Sustain. Comput.* Cham, Switzerland: Springer, 2022, pp. 77–469.

[36] L. Chen et al., "Improve unsupervised learning-based landslides detection by band ratio processing of RGB optical images: A case study on rainfall-induced landslide clusters," *Geomatics, Natural Hazards Risk*, vol. 15, no. 1, Dec. 2024, Art. no. 2363406.

[37] R. Coluzzi et al., "Rapid landslide detection from free optical satellite imagery using a robust change detection technique," *Sci. Rep.*, vol. 15, no. 1, p. 4697, Feb. 2025.

[38] Y. Ling, Y. Wang, Y. Lin, T. On Chan, and J. Awange, "Optimized landslide segmentation from satellite imagery based on multiresolution fusion and attention mechanism," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[39] S. H S, K. Sooda, and B. Karunakara Rai, "EfficientNet-B7 framework for anomaly detection in mammogram images," *Multimedia Tools Appl.*, vol. 84, no. 11, pp. 1–27, May 2024.

[40] S. Dharaneswar and B. P. Santosh Kumar, "Elucidating the novel framework of liver tumour segmentation and classification using improved optimization-assisted EfficientNet B7 learning model," *Biomed. Signal Process. Control*, vol. 100, Feb. 2025, Art. no. 107045.

[41] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.

[42] N. Li, Y. Chen, and D. Zhao, "Dense attention: A densely connected attention mechanism for vision transformer," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2023, pp. 1–8.

[43] X. Zhang et al., "DifUnet++: A satellite images change detection network based on UNet++ and differential pyramid," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[44] P. Shi, X. Xu, X. Fan, X. Yang, and Y. Xin, "LL-UNet++: UNet++ based nested skip connections network for low-light image enhancement," *IEEE Trans. Comput. Imag.*, vol. 10, pp. 510–521, 2024.

[45] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020.

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[46] A. Hou et al., "Brain segmentation based on UNet++ with weighted parameters and convolutional neural network," in *Proc. IEEE Int. Conf. Adv. Electr. Eng. Comput. Appl. (AEECA)*, Aug. 2021, pp. 644–648.
[47] H. Chen et al., "A multi-input channel U-Net landslide detection method fusing SAR multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1215–1232, 2024.
[48] Y. Zhu et al., "Segmentation of mining subsidence areas in D-InSAR interferometric phase images using improved UNet++ network," *J. Appl. Remote Sens.*, vol. 18, no. 3, Sep. 2024, Art. no. 034522.
[49] X. Jin, P. Zhang, Q. Jiang, S. Miao, S. Yao, and W. Zhou, "F-UNet++: Remote sensing image fusion based on multipurpose adaptive shuffle attention and composite multi-input reconstruction network," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–15, 2023.
[50] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.
[51] B. Yu, F. Chen, C. Xu, L. Wang, and N. Wang, "Matrix SegNet: A practical deep learning framework for landslide mapping from images of different areas with different spatial resolutions," *Remote Sens.*, vol. 13, no. 16, p. 3158, Aug. 2021.
[52] H. Yan, A. Khan, A. Jamil, B. Abdeldjalil, T. Saidani, and N. Y. Rebouh, "Deep learning-based spatial prediction of landslide risk in coastal areas using GIS and multicriteria decision making: A DeepLabV3+ approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 15222–15235, 2025.
[53] Z. Li, W. Liu, G. Wu, and S. Yang, "Semantic segmentation of unmanned aerial vehicle image based on resnet-unet," in *Proc. 8th Int. Conf. Intell. Informat. Biomed. Sci. (ICIIBMS)*, Nov. 2023, pp. 295–299.
[54] J. N. Opara, R. Moriwaki, and P. J. Chun, "Automated landslide mapping in Japan using the segformer model: Enhancing accuracy and efficiency in disaster management," *Intell., Inform. Infrastruct.*, vol. 4, no. 2, pp. 75–86, 2023.



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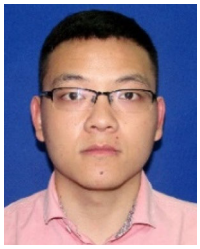
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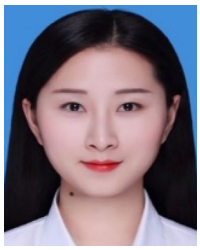
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